Solution Requirements

Date	29 June 2026
Team ID	N/A
Project Name	OptiCrop smart agricultural production optimization engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Farmers and admins can securely log in using username and password	High
2	Authorization levels	Admin manages data; farmers can access crop recommendations only.	High

3	External interfaces	Connects to weather API or	High
		soil database for crop prediction.	
4	Transactions processing	Processes user inputs (soil, temperature, rainfall, humidity) and generates crop recommendations.	High
5	Reporting	Displays recommended crops and prediction results.	Medium
6	Business rules, etc.	Recommendations are based on AI model predictions using user input.	High
7	Compliance to laws or regulations	Protects user data and follows privacy and security guidelines.	Medium

8	*Other*	Allows users to view crop suggestions anytime.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	Crop recommendation should be generated quickly.	Response time < 3 seconds
2	Scalability	Supports many users simultaneously.	Up to 1000 users
3	Security & Data Privacy	Encrypt user data and secure login.	AES-256 encryption
4	Reliability & Availability	System should be available most of the time.	99.9% uptime

5	Usability & Accessibility	Simple, user-friendly interface for farmers.	Easy navigation
6	*Other*	Easy to maintain and update the AI model. High maintainability	High maintainability